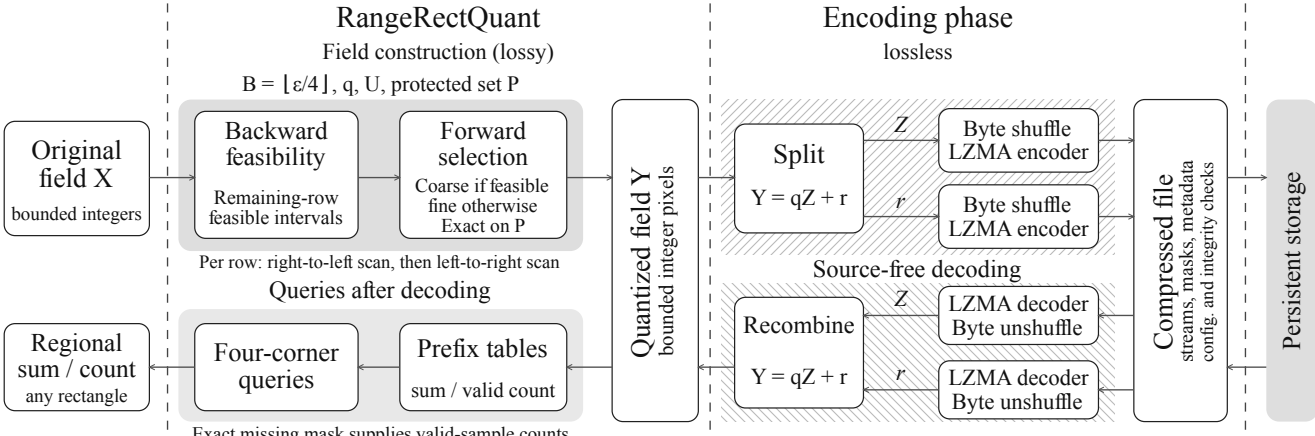

RangeRectQuant
Field construction (lossy)
B = ⌊ε/4⌋, q, U, protected set P
Original field X
bounded integers
Backward feasibility
Remaining-row feasible intervals
Forward selection
Coarse if feasible fine otherwise Exact on P
Per row: right-to-left scan, then left-to-right scan
Quantized field Y
bounded integer pixels
Encoding phase
lossless
Split
Y = qZ + r
Z
r
Byte shuffle LZMA encoder
Byte shuffle LZMA encoder
Source-free decoding
Recombine
Y = qZ + r
LZMA decoder Byte unshuffle
LZMA decoder Byte unshuffle
Compressed file
streams, masks, metadata
config. and integrity checks
Persistent storage
Queries after decoding
Regional sum / count
any rectangle
Four-corner queries
Prefix tables
sum / valid count
Exact missing mask supplies valid-sample counts
0 ≤ Y ≤ U, Y = X on P, |prefix error| ≤ B ⇒ every rectangular sum error ≤ ε